# An Optimality Principle Governing Human Walking

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***Abstract*—In this paper, we investigate different possible strategies underlying the formation of human locomotor trajectories in goal-directed walking. Seven subjects were asked to walk within a motion capture facility from a fixed starting point and direction, and to cross over distant porches for which both position and direction in the room were changed over trials. Stereotyped trajectories were observed in the different subjects. The underlying idea to attack this question has been to relate this problem to an optimal control scheme: the trajectory is chosen according to some optimization principle. This is our basic starting assumption. The subject being viewed as a controlled system, we tried to identify several criteria that could be optimized. Is it the time to perform the trajectory? The length of the path? The minimum jerk along the path?... We found that the variation (time derivative) of the curvature of the locomotor paths is minimized. Moreover, we show that the human locomotor trajectories are well approximated by the geodesics of a differential system minimizing the $L_2$ norm of the control. Such geodesics are made of arcs of clothoids. The clothoid or Cornu spiral is a curve, whose curvature grows with the distance from the origin.**

***Index Terms*—Biological motor systems, optimal control.**

## I. Introduction

The study of sensorimotor control in biological systems has been a major source of inspiration in the always improving quest to better design autonomous machines. This has led roboticists to expand enormously their interaction with the life science community over the last decades. As a result, many exciting developments and novel applications have arisen from the humanoid, the biomedical, and the biomechatronics research areas (among others). The roboticist's traditional emphasis has been on applying the principles underlying complex behaviors of biological systems to implement sophisticated robotic interfaces. On the flip side, the neuroscientists are interested in the tools emerging from the computational approach developed by the roboticists to formalize the knowledge acquired by experimentation in terms of mathematical models. As a consequence of this neurophysiological perspective, appropriate experimental protocols must be defined to exhibit the behavior under study. In much the same way, the motivation of the paper presented here aims to apply a computational approach in the area of movement neuroscience (for a review, see [43]).

The pioneering work of Bernstein to identify the strategies of biological motor control (or control law) suppose a categorization of control models [4]. The first class of models focuses on open-loop control, which plan and execute the motion, ignoring the role of sensory feedback information. The second class of models focuses on closed-loop control to predict and correct deviations away from the current motion execution by online sensory feedback. In addition to what has been stated, and following the Bernstein's approach, different motor levels of description must be taken into account to accomplish a motor task (even in simple tasks like arm motion between two different spatial positions) such as dimensionality (the number of degrees of freedom of the mechanism), redundancy, and the apparent existence of an infinite number of solutions. This has motivated the experimental study of motor patterns in order to find motor invariants in the generation of biological movements. Consequently, many theories of motor commands are based on an optimal control perspective: find a natural optimal performance, like energy consumption, to predict averaged body or limbs' trajectories.

Goal-directed locomotion in humans has mainly been investigated with respect to how different *sensory inputs* are dynamically integrated, facilitating the elaboration of locomotor commands that allow reaching a desired body position in space. Visual, vestibular, and proprioceptive inputs were analyzed during both normal and blindfolded locomotion in order to study how humans could continuously control their trajectories (see [17], and for a review, see [20]). The interaction between the relative motion of the head, the torso, and the eyes has also been studied [18]. However, the principles underlying the generation (or planning) of locomotor trajectories received little attention. Recently, it has been observed that for predefined paths, an inverse relationship between the path geometry (curvature profile) and body kinematics (walking speed) exists [21], [42]. This empirical relation known as *"the power law"* was previously observed in [26] for drawing and handwriting movements. In particular, the so-called one-third power law was consistently reported in different experimental conditions even if no physical reason relating speed and curvature exists (see [21] for a review). Moreover, other studies suggest that the power law seems to be a byproduct of a more complex behavior [34], [38]. Recent approaches based on optimization theory provide optimality principles that encode a cost function to be minimized (for a review, see [39]). These methods are used to predict optimal movements by searching the control law according to some performance criterion.

Different hypotheses, like the maximization of the smoothness [14], [38], have been used for characterizing the production of motor behavior. In [41], the authors proposed a modified

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minimum jerk model accounting for the two-thirds power law. The minimum torque-change model [40] is also a smoothness optimization principle in which dynamics is involved. This model predicts also straight line hand trajectories that correspond to the hand trajectories observed experimentally. The minimum variance model has been proposed for both eye and arm movements [19]. This model suggests that the neural control signal is corrupted by noise. The predicted velocity profiles of eye and arm trajectories account for the speed–accuracy compromise stated by the Fitts' law [15]. Even if these models capture many aspects of observed hand trajectories and the minimum variance model also predicts eye trajectories, they have not been applied, in our knowledge, for locomotor planning. The contribution of this paper is the application of optimal control methods to find the underlying principle explaining the shape of human locomotor trajectories.

When humans walk, the displacement of the limbs, body, and head is coordinated in order to reach a given target. Such coordinations reduce the dimension of the motor space associated with the number of body articulations. We can imagine the possibility that the whole body motion is mainly constructed at the step level by the influence of the leg's movements. The validity of the hypothesis that goal-directed locomotion is planned as a succession of footsteps has been recently discussed by the authors in [22]. We described the spatial and temporal features of the locomotor trajectories when humans perform natural displacements. We argued that goal-directed locomotion may be planned as a whole on the trajectory level rather than successive footsteps. These observations confirm the validity that a human locomotion model can be derived by exclusively looking at the shape of locomotor trajectories and by ignoring all the body biomechanical motor controls generating the motions. Then, we rely on the observation of the geometric shape of the locomotor trajectories in the simple 3-D space of both the position and the direction of the body. These trajectories are the geodesics of the system that we try to identify according to some optimization principle. The reachable space of the proposed control model covers the entire $R^2 \times S^1$ configuration space of the position and the direction of the human body. However, our model accounts only for a part of the human locomotion strategies. This is because both *backward* and *sideway* step motions are not accounted for by our model. In other words, we just focus on forward "natural" locomotion when the goal is defined *in front of* the start configuration.

In Section I, we propose a differential system accounting for human forward locomotion. We overview the validation of the proposed model by an experimental protocol involving seven subjects walking in a motion capture facility. This study has been previously published in [1]. The remaining sections of the current paper aim at explaining the shape of human locomotor trajectories via optimal control.

## II. A Control Model of Human Locomotion

What is the differential controlled system that accounts for the human locomotion at best? What experimental protocol validates the model? The first stage of this work has been based

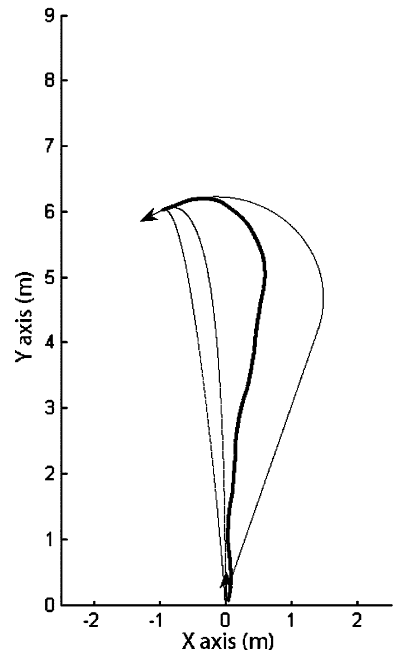



Fig. 1. Among four "possible" trajectories reaching the same goal, the subject has chosen the bold one. Why?

on a simple statement: "the natural way for walking is to put one foot *in front of* the other and to repeat this action." Indeed, "*in front of*" means that the direction of the motion is given by the direction of the body: it implies a coupling between the direction of the body and the tangent to the trajectory. This is a differential nonintegrable coupling known as being nonholonomic.[1] The first part of this research has been to prove this statement and to provide a first control system that accounts for the locomotor trajectories. We follow a methodology based on a geometric study of the accessibility domain of the forward locomotor trajectories.

First of all, we stated the problem within the 3-D space of body position and direction, giving rise to the question illustrated in Fig. 1. We restrict the study to the "natural" forward locomotion with nominal speed. The model we study should be valid for all possible intentional goals reachable by a forward walk.[2] We exclude from the study the goals located behind the starting position and the goals requiring side walk steps.[3] Then, we defined an experimental protocol accounting for the intentional trajectories whose goals are defined both in position and direction. Because the objective was to cover at best the 3-D accessibility region, we sampled the domain with 480 points defined by 40 positions on a 2-D grid (within a 5 m $\times$ 9 m

[1]Nonholonomy is a classical concept from mechanics that has been very fruitful in mobile robotics over the past 20 years.

[2]In an empty space, any goal, even located behind the starting position may be reachable by a forward walking. However, this is not the "natural" way to do so.

[3]This is an important assumption: it is related to the accessibility space of a control system. Here, we reasonably assume that the accessibility domain of the forward locomotion is a kind of a 3-D cone approximated by the accessibility domain that we consider in the protocol. Drawing the "exact" frontiers of the forward locomotion accessibility domain is typically a topic for future work opened by this study.

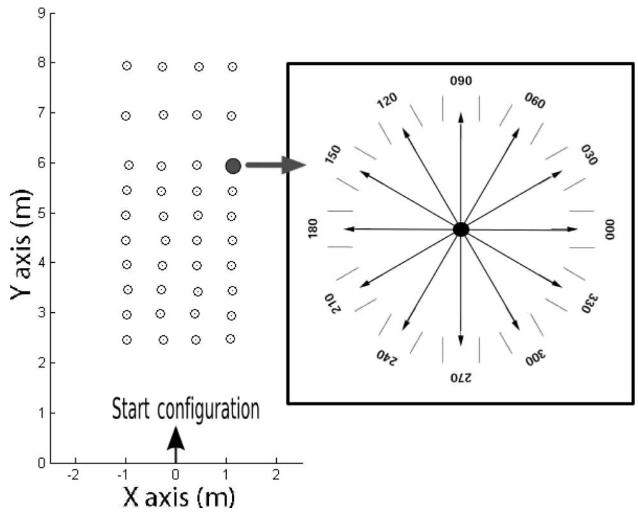



Fig. 2. It shows all the final configurations considered in the study. We sampled a region of the gymnasium with 480 points defined by 40 positions on floor (within a 5 m $\times$ 9 m rectangle) and 12 directions each. More precisely, the final direction varied from $-\pi$ to $\pi$ in intervals of $\frac{\pi}{6}$ at each final position.

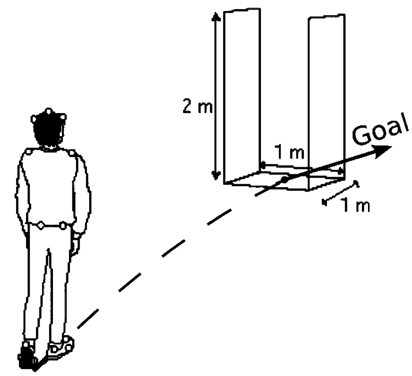



Fig. 3. Porch used in the experiments. The starting position was always the same while the goal was randomly selected.

rectangle) and 12 directions each (see Fig. 2, all the pictures are displayed in the same frame). The starting position was always the same. Locomotor trajectories were recorded in a large gymnasium in seven normal healthy males who volunteered for participation in the experiments. Their ages, heights, and weights ranged from 26 to 29 years, from 1.75 to 1.80 m, and from 68 to 80 kg, respectively. One subject performed all the 480 trajectories while the other six performed only a subset of them chosen at random. Subjects walked from the same initial configuration to a randomly selected final configuration. The target consisted of a porch that could be rotated around a fixed point to indicate the desired final direction (see Fig. 3). The subjects were instructed to freely cross over this porch without any spatial constraints relative to the path they might take. They were allowed to choose their natural walking speed to perform the task. We used motion capture technology to record more than 1500 real trajectories (see Fig. 4). Subjects were equipped with 34 light reflective markers located on their bodies. Among the markers directly used for the analysis, the torso position (middle point $(x_T, y_T)$ between the left and the right shoulders) and direction $\varphi_T$ were found to obey a simple nonholonomic system given by

$$\begin{pmatrix} \dot{x_T} \\ \dot{y_T} \\ \dot{\varphi_T} \end{pmatrix} = \begin{pmatrix} \cos\varphi_T \\ \sin\varphi_T \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u_2 \tag{1}$$

where the control inputs $u_1$ and $u_2$ are the linear and angular velocities, respectively. It is known that the following equation

$$\dot{y_T}\cos\varphi_T - \dot{x_T}\sin\varphi_T = 0 \tag{2}$$

defines a nonintegrable 2-D distribution in the 3-D manifold $R^2 \times S^1$ gathering all the configurations $(x_T, y_T, \varphi_T)$: the coupling between the position and the direction is said to be a nonholonomic constraint. Both linear and angular velocities appear as the only two controls that perfectly define the shape of the paths in the 3-D manifold $R^2 \times S^1$. We then used the torso trajectories for the second stage of our analysis.

## III. Understanding the Geometric Shape of Locomotor Trajectories via Optimal Control

Our approach aims at explaining the shape of the locomotor trajectories by optimal control. By nature, the validation of the control model that we are looking for should be done by comparing the trajectories simulated from the proposed model with a set of observed trajectories. We first have to find a control system that "reasonably" accounts for the human locomotion. Then, we have to find an optimal cost that "reasonably" accounts for the shape of the trajectories. "Reasonably" means that we want a human locomotion model that applies as closely as possible to a set of observed data: the "proofs" will come from statistical analysis. Our approach underlies the following problems:

1) A data basis of trajectories being given, we should find a control model with an associated optimal cost. The inputs of a standard optimal control problem are a model and an associated cost function. The outputs are the optimal trajectories. Here, we assume that the observed trajectories are optimal and we should find the corresponding model and cost. This problem is then viewed as an *inverse* optimal control problem (i.e., given a control system, the reachable space, and optimal trajectories, which is the optimal criterion that steers the system?) [29], [30]. Unfortunately, it is not evident to answer this question in the context of motor control even if it could be more useful. Nevertheless, some tools of optimal control theory are still useful to characterize optimal trajectories that verify the nonholonomic constraints.
2) It also pretends to account for a "*global*" point of view while most of the theoretical results hold only locally.

Our work takes advantage of both analytical and numerical optimal control approaches. First, we apply analytical methods to characterize locally the geometric shape of the geodesics. Then, we apply a numerical optimization algorithm to validate the following hypothesis: locomotor trajectories are well approximated by the optimal solutions of a dynamic extension of a simple unicycle control model. The validation method

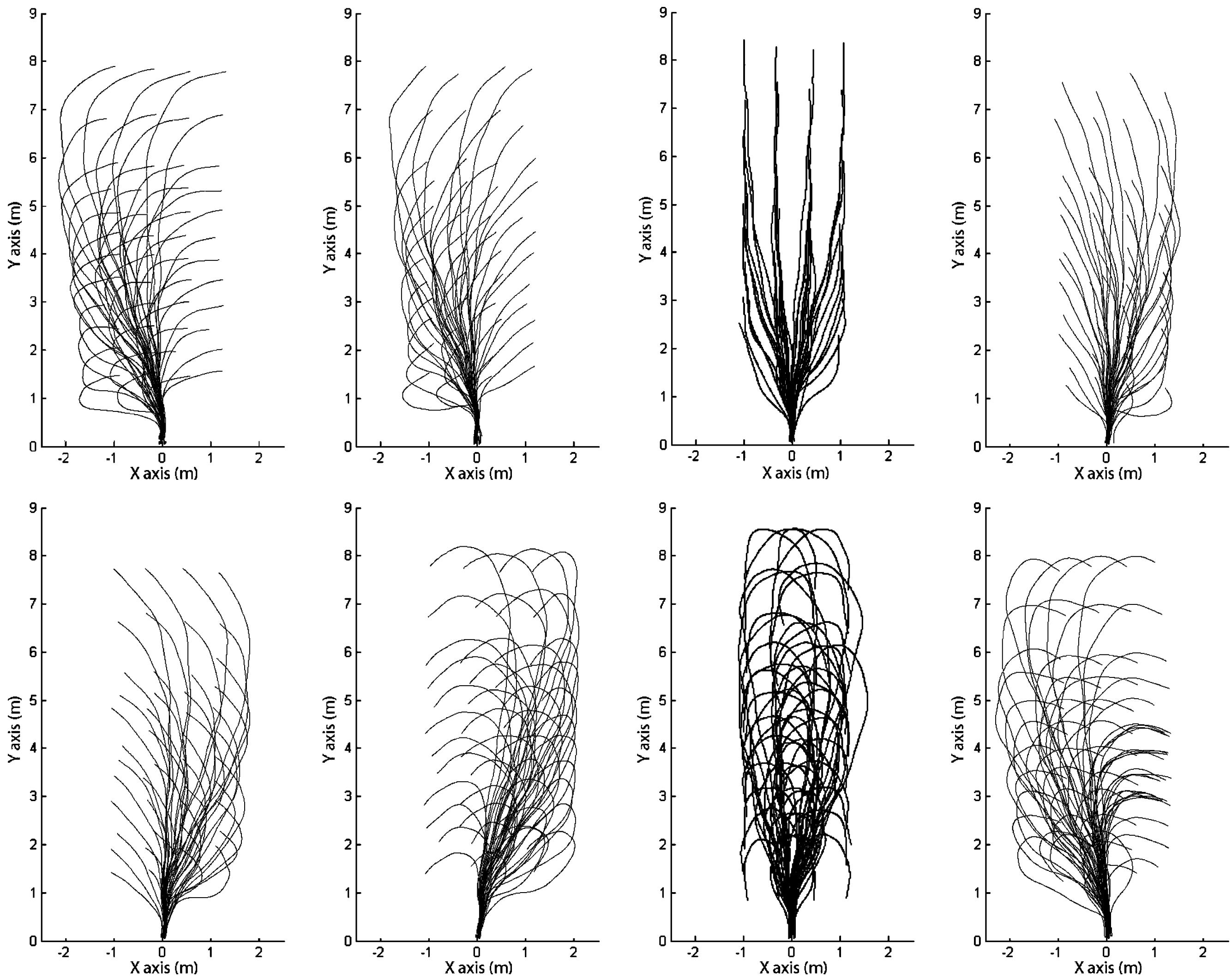



Fig. 4. Some examples of real trajectories with the same final direction. The trajectories are those of the torso. Each picture shows all real trajectories where the final direction is 0, $\frac{\pi}{6}$, $\frac{\pi}{2}$, $\frac{2\pi}{3}$, $\frac{5\pi}{6}$, $\frac{4\pi}{3}$, $\frac{3\pi}{2}$, and $\frac{11\pi}{6}$, respectively.

consists in comparing the optimal trajectories of the system with the trajectories of the data basis.

### A. Model and PMP Analysis

The neuroscience approaches in modeling human motion pointed out the critical role of the curvature [21], [26], [38], [41], [42]. To prevent curvature discontinuities, we propose to make the curvature a variable of the system. We then consider a dynamic extension of System (1) given by

$$\begin{pmatrix} \dot{x_T} \\ \dot{y_T} \\ \dot{\varphi_T} \\ \dot{\kappa_T} \end{pmatrix} = \begin{pmatrix} \cos\varphi_T \\ \sin\varphi_T \\ \kappa_T \\ 0 \end{pmatrix} u_1 + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} u_2 \tag{3}$$

where $u_1$ is the linear velocity and $u_2$ is the control of the time derivative of the curvature. By applying the lie algebra rank condition (LARC [37]) to this control system, it is proved to be controllable. This means that any configuration can be reached from any other one (see [27] and [8]).

Assuming $u_1 \in [a, b]$ with $a > 0$ (forward motion) and $u_2 \in [-c, c]$, we consider the cost function

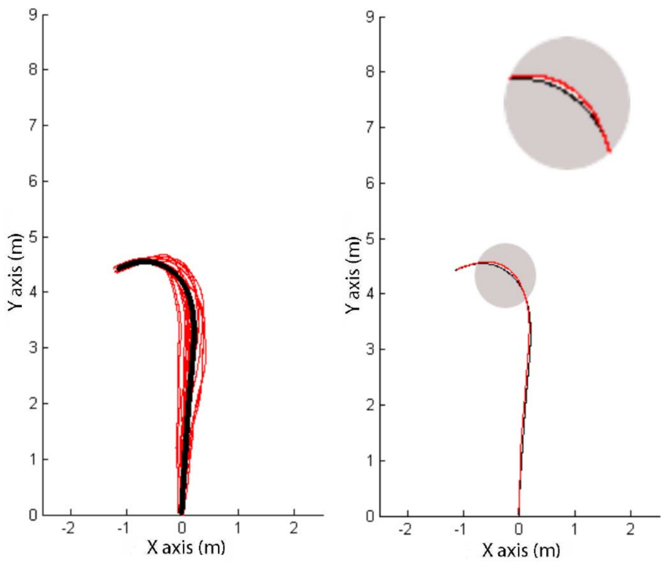
$$J = \frac{1}{2}\int_0^T < (u(\tau), u(\tau)) > d\tau. \tag{4}$$

Applying the maximum principle [31], we found that the optimal trajectories verify locally that $u_1^2 + u_2^2$ should be constant (see Appendix A). The result is not surprising (see [33]).[4] It has not been possible to deduce more information from the maximum principle. Then, we fell back on the numerical optimization algorithms.

### B. Numerical Analysis

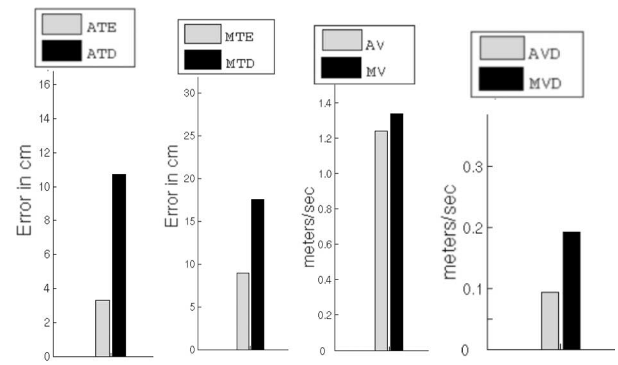
The numerical algorithm that we used has been proposed in [13] (see Appendix B). We first applied the numerical algorithm to System (1) for all the trajectories performed by the

[4]Note: However the result only holds if $u_2 \not\equiv 0$. In the numerical analysis that we performed, $u_2$ is never zero over a nonempty time interval for the considered reachable space. The case $u_2 \equiv 0$ (arcs of a circle and straight line segments) may appear for long range paths.




(a) (b)

Fig. 5. Representative example of statistics computed over the entire movement for the same initial and final configurations. Each subject has done three trials. (a) Shows the mean trajectory (bold) from the real ones (thin). All these trajectories correspond to the same initial and final configurations. It illustrates the variability pattern. (b) Shows the mean trajectory (thin) and the predicted optimal control effort trajectory linking the same initial and final configurations (bold).




(a) (b) (c) (d)

Fig. 6. Accuracy of the model is also supported by the fact that the predicted trajectory is closer to the corresponding recorded trajectory than the trial-by-trial variability of recorded trajectories. (a) Shows the comparison between the averaged trajectory errors (ATE) and the averaged trajectory deviations (ATD). (b) Shows the comparison between the maximal trajectory errors (MTE) and the maximal trajectory deviations (MTD). (c) Shows the averaged and maximal velocity profile (AV) and (MV) of real locomotor trajectories. (d) Shows the comparison between the averaged and maximal velocity deviations (AVD) and (MVD) of real locomotor trajectories.

seven subjects. The results were not satisfactory. This has been the motivation to envisage the differential system with inertial control law with two control inputs: the linear velocity and the derivative of the curvature. The system is given by (3). To validate the model, we compared the predicted trajectories to the recorded ones performed by the seven subjects. The cost function considered was the control effort expended.

To examine the trial-to-trial and subject-to-subject variability for the same plan, we considered the geometric mean as the statistical measure. Because each subject did not spend the same time performing the task (even from trial-to-trial, the duration of the motion is different), we used the duration of the predicted trajectory as the reference in order to compare all recorded trajectories with respect to the predicted one. We then computed the mean at instant $\tau \in [0, T]$ [see Fig. 5(a)]. In this way, it appears that the geometric shape of the averaged trajectory (from experimental data) is highly close to the optimal control effort trajectory (predicted). It can be seen in Fig. 5(b) that the prediction gives a reasonable fit of the experimantal data.

To measure how well the model predicts locomotor trajectories, we compute the difference between both trajectories at the instant $\tau$. To do that, we define the trajectory error TE such as

$$\mathrm{TE}(\tau) = \sqrt{(x_r(\tau) - x_p(\tau))^2 + (y_r(\tau) - y_p(\tau))^2} \qquad (5)$$

where $(x_r(\tau), y_r(\tau))$ and $(x_p(\tau), y_p(\tau))$ are the positions at the instant $\tau$ of the recorded and the predicted trajectories, respectively. Then, we compute the averaged and the maximal trajectory errors

$$\mathrm{ATE} = \int_{\tau\in[0,T]} \mathrm{TE}(\tau)\, d\tau$$
$$\mathrm{MTE} = \max_{\tau\in[0,T]} \mathrm{TE}(\tau). \qquad (6)$$

These two quantities indicate the similarity between the predicted and the recorded trajectories. Thus, small values of ATE and MTE mean that the similarity degree is high between both trajectories.

This procedure has been executed on the 1560 trajectories performed by the seven subjects. It is interesting to note that the model approximates 90% of trajectories with an average error <10 cm and a maximal error <20 cm.

The accuracy of the model is also supported by the fact that ATE and MTE are always lower than the averaged ATD and the maximal MTD trajectory deviations for the subset of trajectories performed by all the subjects (see Fig. 6(a) and (b)). These deviations are computed between the recorded and the mean trajectories corresponding to the same target. In other words, the predicted trajectory is always inside the area defined by the trial-to-trial variability of recorded trajectories. Consequently, this study proves that:

1) the locomotor trajectories are well approximated by the optimal solutions of a dynamic extension of a simple unicycle model;
2) the locomotor trajectories minimize the time derivative of the curvature.

In addition to the path geometric variability, we performed the same procedure for the trajectory kinematic attributes. To quantify the variability of the linear velocity profile among subjects and trials, we computed the mean and the maximal linear velocity profiles AV and MV, respectively. Then, we calculated the averaged AVD and the maximal MVD linear velocity deviations [see Fig. 6(c) and (d)]. The statistical analysis shows that $u_1$ control remains "reasonably" constant over the whole interval of time (we should keep in mind that the subjects were asked to enter the room by the starting configuration while not stopping at the goal).

According to the PMP analysis of our optimization problem, $u_1^2 + u_2^2$ should be constant (see Section III-A and Appendix A). Then, we can deduce that $u_2$ should be a piecewise constant function. Since $\dot{\kappa}_T = u_2$ and considering that $u_2 = cu_1$, by integration we obtain that $\kappa_T = cs + \kappa_0$, where $\kappa_0$ is the initial curvature, $s = u_1\tau$ and $c$ a constant. A curve followed at constant velocity while linearly increasing or decreasing the curvature is known as being a clothoid. Therefore, clothoid arcs are a good approximation of locomotor trajectories. For System (3) moving on a curve at constant velocity, the only acceleration is the centripetal acceleration given by $a_c = v^2\kappa$. The *jerk*, which is the time derivative of the acceleration, is given by

$$j_c = v^2\dot{\kappa} \quad or \quad j_c = v^3\frac{d\kappa}{ds} \tag{7}$$

where $s$ is the arc length. Since a pair of clothoid arcs has a triangular curvature pattern, and the sides of the triangle have slopes equal to $\pm j_c/v^3$, it corresponds to the minimum length curve under a peak-jerk constraint.

Fig. 7 shows a representative real trajectory performed by one subject and the optimal control effort trajectory linking the same initial and final configurations. The real trajectory has been filtered to illustrate the comparison between the control inputs extracted from the real (filtered) trajectory and the computed optimal control inputs. Fig. 8 shows some examples of the behavior of the real and predicted trajectories by translating the final position over both: the vertical and the horizontal axes with a fixed final direction. Fig. 9 shows the symmetric recorded and predicted trajectories. The final direction varies in intervals of $\frac{\pi}{6}$. Fig. 10(a) shows some examples of real and predicted trajectories for a fixed final position.

**Note:** Clothoids are used for trajectory smoothening in robotics [16], [25], [35] (see [10], [11], and [23] on motion control for autonomous robots). A clothoid, also known as a Cornu spiral, is a curve along which the curvature $\kappa$ depends linearly on the arc length and varies continuously from $-\infty$ to $+\infty$. Their equation is $\kappa = cs + \kappa_0$, where $s$ is the arc length, $\kappa_0$ the initial curvature, and $c$ is the sharpness or a constant characterizing the shape of the clothoid (see Fig. 11). Clothoid curves correspond to the minimum-length continuous curvature paths under a centripetal peak-jerk constraint (i.e., the jerk is proportional to the slope of the curvature pattern). Nevertheless, other types of continuous curvature curves for mobile robots have been proposed. A different type from the one first considered is the *cubic spiral* [24]. It is a curve along which the curvature is a cubic function on the arc length. This type of curve corresponds to minimizing the integral of the square of the centripetal jerk (i.e., it has a curvature function that is parabolic in shape). Moreover, a generalization of the clothoids and cubic spiral curves is the *intrinsic spline* whose curvature is a polynomial function of the arc length [9].

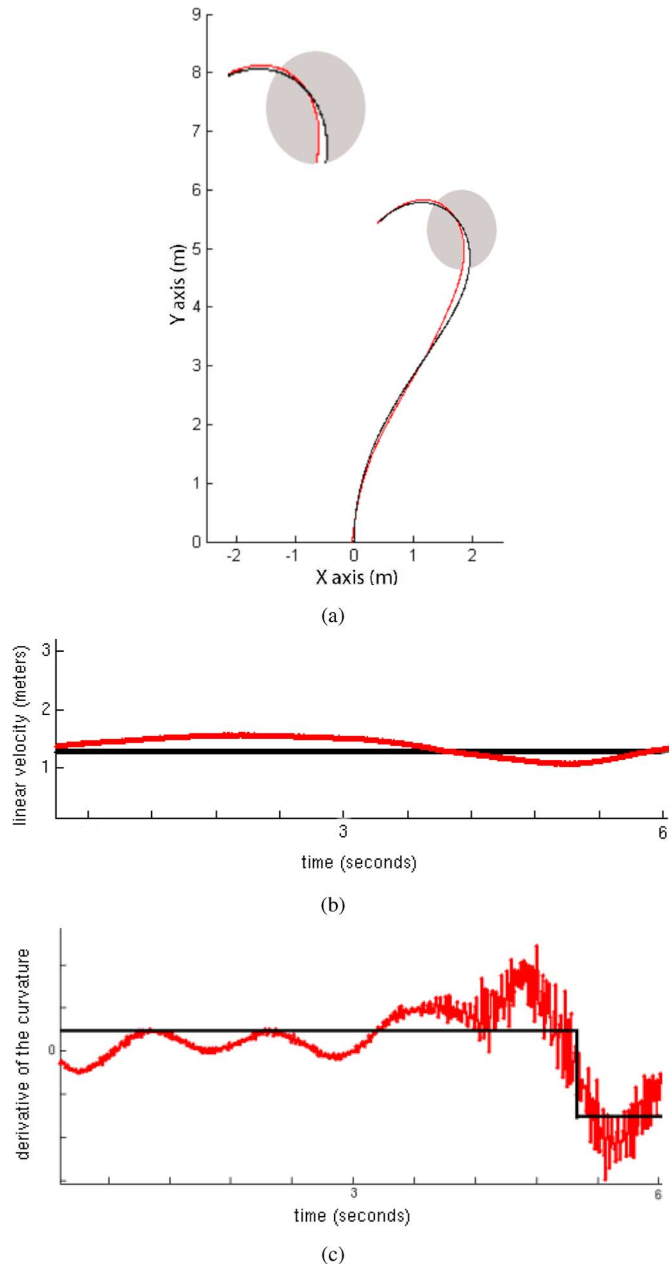
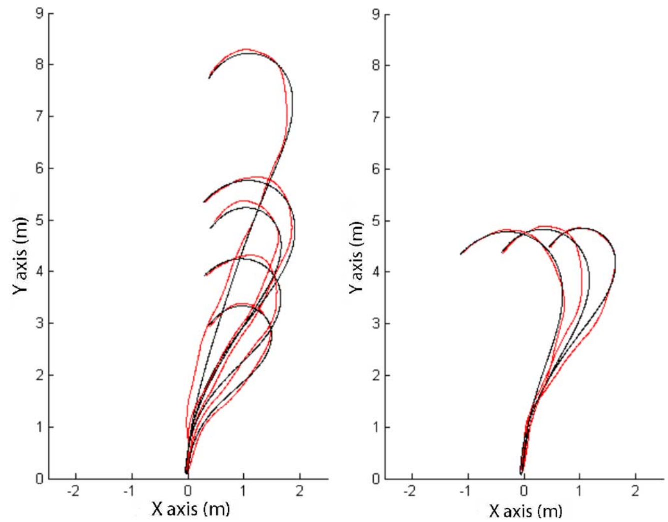



Fig. 7. Representative examples of comparisons between real (thin) and predicted (bold) locomotor trajectories. (a) Shows a real trajectory performed by the first subject (thin) and the predicted optimal control effort trajectory linking the same initial and final configurations (bold). (b) and (c) Show the comparison between the control inputs extracted from the real trajectory (filtered) of (a) and the computed optimal control inputs extracted from the optimal control effort trajectory.

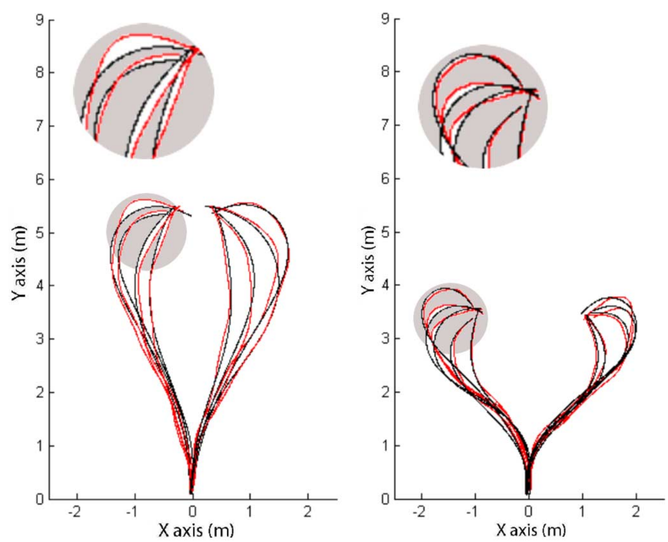
## IV. Discussion

In the first part of this paper, we have shown that the forward human locomotion, represented by the torso position and direction, obeys the motion of a nonholonomic system with linear and angular velocity inputs. In the second part, we were able to predict more than 90% of the 1560 trajectories recorded from seven subjects during walking tasks with a $<$10 cm accuracy. We have implemented a numerical optimization algorithm to validate that the locomotor trajectories are well approximated by the



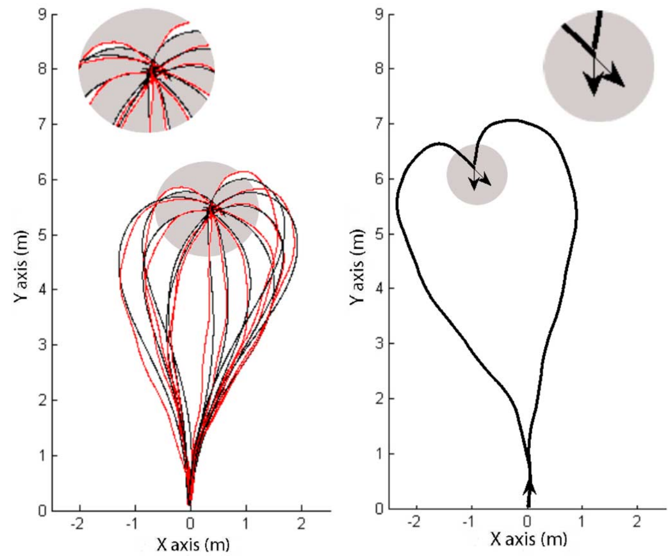

(a) (b)

Fig. 8. Representative examples of comparisons between real (thin) and predicted (bold) locomotor trajectories. (a) Shows the behavior of the real and predicted trajectories by translating the final position in the vertical axis with a fixed final direction. (b) Shows the behavior of the real and predicted trajectories by translating the final position in the horizontal axis with a fixed final direction.




(a) (b)

Fig. 9. Representative examples of comparisons between recorded (thin) and integrated (bold) locomotor trajectories. (a) and (b) Show the symmetric recorded and predicted trajectories. The final direction varies in intervals of $\pi/6$.




(a) (b)

Fig. 10. Representative examples of comparisons between real (thin) and predicted (bold) locomotor trajectories. (a) Shows some examples of real and predicted trajectories for a fixed final position. The final direction varies in intervals of $\frac{\pi}{6}$ (b) Illustrates the decision processes of the natural question: Should I reach the goal by the left or by the right side when the final position is not in front of the initial configuration?

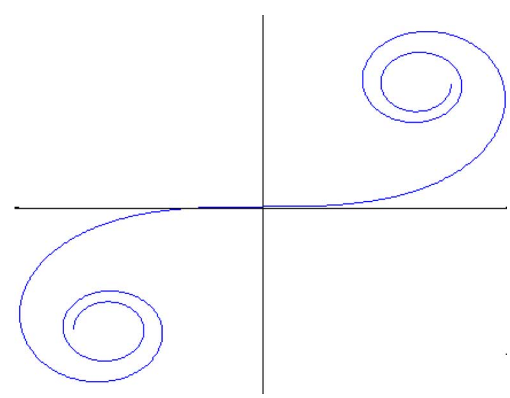


Fig. 11. Cornu spiral or clothoid is a curve whose curvature grows with the distance from the origin.

optimal solutions of a dynamic extension of the unicycle model. The criterion to be minimized has been the time derivative of the curvature. Using an analytical optimal control approach, we have locally characterized the geometric shape of the geodesics. The computation for finding the number of concatenated arcs of clothoids (switching points) has been reported in [28].

As a conclusion, let us pinpoint an issue that opens challenging perspectives. Figs. 8 and 9 illustrate a natural intuition: when the goal position varies slightly (with fixed direction, Fig. 8) and when the goal direction varies slightly (with fixed positions, Fig. 9) the resulting trajectories are obtained by continuous reshaping. This means that close goals give rise to close trajectories. Let us consider now the case illustrated in Fig. 10(b). The goal position of both the trajectories is the same. Both goal directions differ slightly while both trajectories completely differ. This case contradicts the previous intuition. This behavior occurs only in some special cases. It arises around points reachable by two distinct trajectories with exactly the same cost. Now the question is: how does the brain identify such cases?

The study of the geometry of nonholonomic trajectories is part of a mathematical domain of differential geometry known as sub-Riemannian geometry [3], [5]. For very few systems, it is possible to compute exactly the locus of goal points reachable by exactly two distinct trajectories (see, for instance, [36] for the case of the car-like robot). This locus is known as the so-called *cut-locus* in sub-Riemannian geometry. However, methods to compute such a locus for any nonholonomic system are out of the scope of the current state of the art in differential geometry. In [28], the authors provide the *cut-locus of the human locomotion* by numerical methods.

## Appendix A

Here we consider the problem of steering the System (3) from an initial state $q(0) = q_0$ to a final state $q(T) = q_f$ minimizing the $L_2$ norm of the control given by (4) which corresponds to the least-squares optimal control problem. To find a set of control inputs $u(\tau) \in R^m, \tau \in [0, T]$, which minimizes the cost $J$ and steers the system from $q_0$ to $q_f$, we apply the Pontryagin's maximum principle (PMP). The PMP states that if $u$ is an admissible control, then $u(\tau)$ and the trajectory $q(\tau)$ are optimal if there exists a nonzero continuous vector function $\psi$ associated to $q(\tau)$ such that $u(\tau)$ maximizes the Hamiltonian function for all $\tau \in [0, T]$. It should be emphasized that the PMP establishes a set of necessary, but not sufficient, conditions for optimality. The Hamiltonian $H$ is defined by

$$\begin{aligned} H = &\frac{1}{2}\left(u_1^2 + u_2^2\right) + \psi_1 \cos\varphi_T u_1 \\ &+ \psi_2 \sin\varphi_T u_1 + \psi_3 \kappa_T u_1 + \psi_4 u_2. \end{aligned}$$

To determine the control that minimizes $H$, it is necessary that

$$\frac{\partial H}{\partial u} = 0 \begin{cases} u_1 + \psi_1 \cos\varphi_T + \psi_2 \sin\varphi_T + \psi_3 \kappa & = 0 \\ u_2 + \psi_4 & = 0. \end{cases}$$

So, we have the adjoint system $\dot{\psi} = -\frac{\partial H}{\partial q}$ (for every $\tau \in [0, T]$)

$$\dot{\psi} = \begin{cases} \dot{\psi}_1 = 0 \\ \dot{\psi}_2 = 0 \\ \dot{\psi}_3 = \psi_1 \sin\varphi_T u_1 - \psi_2 \cos\varphi_T u_1 \\ \dot{\psi}_4 = -\psi_3 u_1. \end{cases}$$

By differentiating the optimal controls,[5] we obtain the following expressions:

$$\begin{aligned} \dot{u}_1 &= -\psi_3 u_2 \\ \dot{u}_2 &= \psi_3 u_1. \end{aligned}$$

Hence, we get

$$\begin{aligned} u_2 \dot{u}_2 &= -u_1 \dot{u}_1 \\ u_1^2 + u_2^2 &= \text{constant}. \end{aligned}$$

This result is not surprising. The general case was proved by Sastry and Montgomery [33].

## Appendix B

Generally, it is difficult to find the solution of the optimal steering of nonholonomic systems, the only possibility is to rely on numerical methods.[6] We describe here the method developed by Fernandes *et al.* [13].

Let us consider the dynamical System (3) together with the cost function given by (4). Denoting by $\{\mathbf{e}_k\}_{k=1}^{\infty}$ an orthonormal basis for $L_2([0, T])$ and considering a continuous and piecewise $C^1$ control law $u$ defined over $[0, T]$, we may write a function $u \in L_2([0, T])$ in terms of a Fourier basis

$$u = \sum_{k=1}^{\infty} \left(\alpha_k \mathbf{e}_k^{i\frac{2k\pi t}{T}} + \beta_k \mathbf{e}_k^{-i\frac{2k\pi t}{T}}\right).$$

Then, $u$ can be approximated by truncating its expansion up to some rank $N$. The new control law $u$ and the objective function $J$ are then expressed as

$$u = \sum_{k=1}^{N} \alpha_k \mathbf{e}_k \Longrightarrow J \simeq \sum_{k=1}^{N} |\alpha_k|^2$$

where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_N) \in R^N$ is to be determined. The configuration $q(T)$ is the solution at time $T$ applying the control law $u$. Clearly, $q(T)$ appears as a function $f(\alpha)$ from $R^N$ to $R^n$. In order to steer the system to $q_f$, an additional term must be added to the cost function

$$J(\alpha) = \sum_{k=1}^{N} |\alpha_k|^2 + \gamma \|f(\alpha) - q_f\|^2$$

where $q(T) = f(\alpha)$ and $\gamma$ is a tuning parameter during the optimization. It is proved that the solution of the new finite-dimensional problem converges to the exact solution as $N$ and $\gamma$ go to infinity [13].

The new optimization problem becomes: given a fixed time $T$ and $q_0, q_f$, find $\alpha \in R^N$ such that the cost function $J(\alpha)$ is minimized. In other words, this approach will give us near-optimal paths. Because $f(\alpha)$ is in most of the cases not known, we should use numerical integration to obtain $f(\alpha)$ and its Jacobian $\frac{\partial f}{\partial \alpha} \in R^{n \times N}$. Thus, let us rewrite the System (3) such that $\forall(\tau, \alpha) \in [0, T] \times [0, +\infty)$

$$\frac{\partial q}{\partial \tau}(\tau, \alpha) = \sum_{i=1}^{m} u_i(\tau, \alpha) B_i(q(\tau, \alpha)) \tag{8}$$

where the columns $B_i(q(\tau, \alpha))$ are the control vector fields. By differentiating (8), we get

$$\frac{\partial^2 q}{\partial \tau \partial \alpha}(\tau, \alpha) = A(\tau, \alpha) Y(\tau, \alpha) + B(\tau, \alpha) \mathbf{e}(\tau, \alpha) \tag{9}$$

where $Y(\tau, \alpha) = \frac{\partial q}{\partial \alpha}(\tau, \alpha)$ and $\mathbf{e}(\tau, \alpha) = \frac{\partial \mathbf{u}}{\partial \alpha}(\tau, \alpha)$ are vector-valued functions. $A(\tau, \alpha)$ is the following $n \times n$ matrix:

$$A(\tau, \alpha) = \sum_{i=1}^{m} \left(u_i(\tau, \alpha) \frac{\partial B_i}{\partial q}(q(\tau, \alpha))\right)$$

and $B(\tau, \alpha)$ is an $n \times m$ matrix. Therefore, the System (9) is, in fact, the linearized system of (8) about $\tau \rightarrow q(\tau, \alpha)$.

[5] We suppose that $u_1$ and $u_2$ are piecewise $C^2$, which is a reasonable assumption.

[6] Few special cases similar to ours have been solved analytically: they deal with the computation of the shortest paths for Dubins's car [12], Reeds and Shepp's car [32], and some extensions of them [6]–[8].

Using the aforesaid equations, $f(\alpha)$ and $\frac{\partial f}{\partial \alpha}$ are obtained by evaluating (8) and (9) at $\tau = T$, respectively. Applying the optimization method to System (3), the matrices $A(\tau)$ and $B(\tau)$ have the following structures:

$$A(\tau) = \begin{pmatrix} 0 & 0 & -u_1 \sin\varphi_T & 0 \\ 0 & 0 & u_1 \cos\varphi_T & 0 \\ 0 & 0 & 0 & u_1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$B(\tau) = \begin{pmatrix} \cos\varphi_T & 0 \\ \sin\varphi_T & 0 \\ \kappa_T & 0 \\ 0 & 1 \end{pmatrix}.$$

We also need to represent the control inputs $u$ over the basis of $N$ vectors given by the following functions:

$$\mathbf{e}_1(\tau) = (1,0)^t, \qquad \mathbf{e}_2(\tau) = (0,1)^t$$

$$\mathbf{e}_3(\tau) = \left(\cos\left(\frac{2\pi\tau}{T}\right), 0\right)^t, \quad \mathbf{e}_4(\tau) = \left(0, \cos\left(\frac{2\pi\tau}{T}\right)\right)^t$$

$$\mathbf{e}_5(\tau) = \left(\sin\left(\frac{2\pi\tau}{T}\right), 0\right)^t, \quad \mathbf{e}_6(\tau) = \left(0, \sin\left(\frac{2\pi\tau}{T}\right)\right)^t$$

$$\vdots \qquad\qquad\qquad \vdots$$

$$\mathbf{e}_{N-3}(\tau) = \left(\cos\left(\frac{2N\pi\tau}{T}\right), 0\right)^t, \quad \mathbf{e}_{N-2}(\tau) = \left(0, \cos\left(\frac{2N\pi\tau}{T}\right)\right)^t$$

$$\mathbf{e}_{N-1}(\tau) = \left(\sin\left(\frac{2N\pi\tau}{T}\right), 0\right)^t, \quad \mathbf{e}_N(\tau) = \left(0, \sin\left(\frac{2N\pi\tau}{T}\right)\right)^t.$$

Let us define by $\Phi$ the $m \times N$ matrix whose columns are the basis elements as

$$\Phi = \begin{pmatrix} \mathbf{e}_1(\tau) & \mathbf{e}_2(\tau) & \ldots & \mathbf{e}_{N-1}(\tau) & \mathbf{e}_N(\tau) \end{pmatrix} \tag{10}$$

where $u = \Phi\alpha$. To compute a solution of the problem, a variation of the Newton's algorithm to update $\alpha$ is used.

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